

Command Relations

- what are they? grammar
- why do we have **two** command relations? syntax-semantics
- Are they **free to vary** among themselves? typology

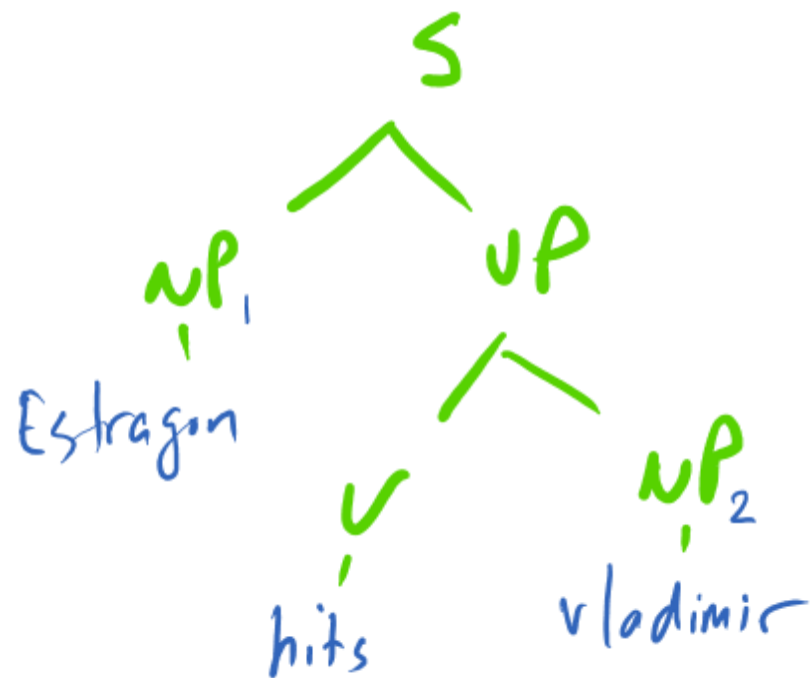
A commands B in general means:

- A can see $\left\{ \begin{array}{l} \text{what happens to} \\ \text{the information in} \end{array} \right\} B$.

- And the inverse does not hold.
- B cannot do these to A.

Syntactic command relations

C-command of prepositionalism



C-commands	
NP ₁	VP
	V
	NP ₂
VP	NP ₁
V	NP ₂
NP ₂	V

S-command of Monodic grammar (Bošković 2005)

S-command	
NP ₁	NP ₂

Because this entire structure can be captured by one function

hits :: (S/NP₁)/NP₂

- S-command has much stricter admissibility.

- C-command must serve all command-related asymmetries, including

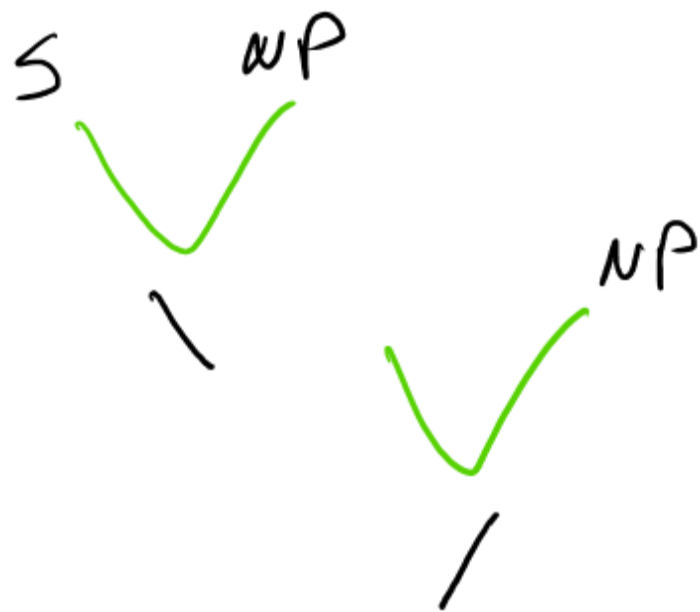
- binding
- constituency
- agreement
- grammatical relations
- scope (quantifiers)

- S-command is syntactic command; it serves syntactic relations

- constituency
- agreement (including its scope)
- grammatical relations

- Binding and quantifier scope is semantically determined syntactically transmitted

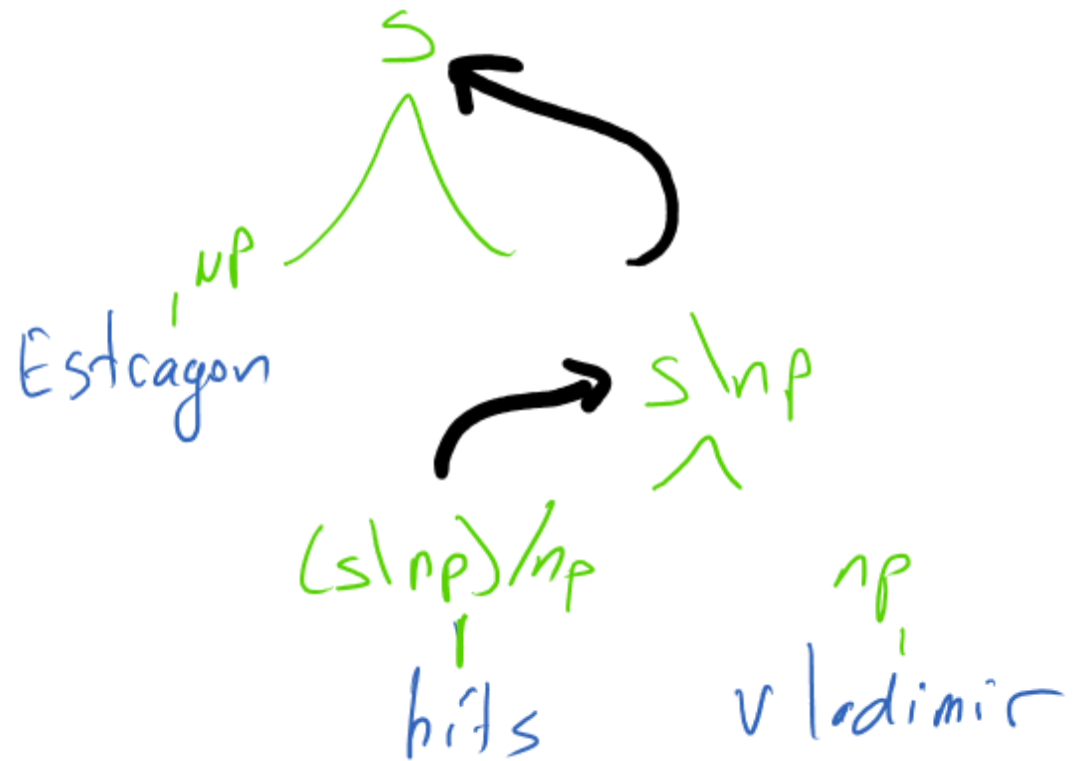
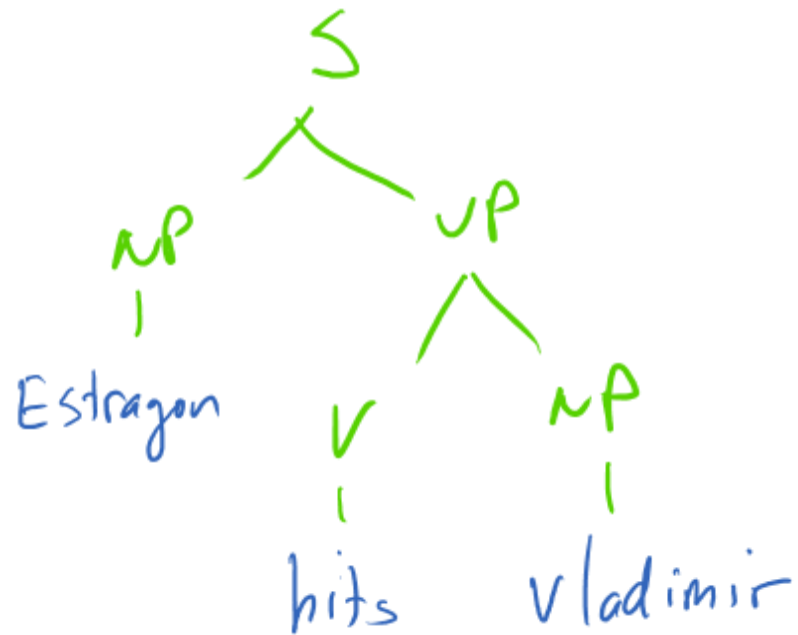
Where is s-command in (slup) np?



s-command
NP₁ | NP₂

S-command is a "later-than" relation. It MUST be a relation between arguments, not any phrase label.

The phrase structure on the left is determinable by a single function using S-command; compare C-command.



Can semantic command be driven by the same function?

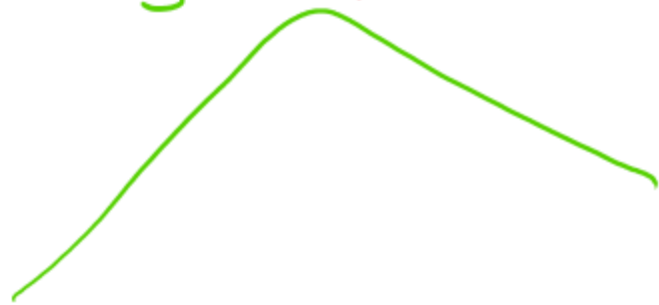
— called 1-command in Bozskis (2025).

hits :: (S \ np) / np : $\lambda x. \lambda y. \text{hit}' x y$



Combining s-command and l-command

$s: \text{hit } v' e'$



$np: e'$

Estragon

$(s \backslash np): \lambda y. \text{hit}' v' y$



$(s \backslash np) / np: \lambda x. \lambda y. \text{hit}' x y$

hits

$np: v'$

Vladimir

$$\begin{aligned}
 (\lambda x. \lambda y. \text{hit}' x y) v' &= \lambda y. \text{hit}' v' y \\
 (\lambda y. \text{hit}' v' y) e' &= \text{hit}' v' e'
 \end{aligned}$$

When the function is more complex, we can
see s-command and l-command doing their
own work TOGETHER without external help:

persuades :: (s \ np) / vp / np : $\lambda x. \lambda p. \lambda y.$
persuade' (px) xy

Compare

expects :: (s \ np) / vp / np : $\lambda x. \lambda p. \lambda y.$
expect' (px) y

- Categorical typology $\approx$ possible
command
variations

- Variants: substantive categories that differ
intra-linguistically and cross-linguistically.

- Invariants: structure of analysis
(for monadic grammar, function
composition).

Q: Are s-command and l-command free to vary within themselves?

A: We need to look at world's languages for that

- This is an empirically testable question.
(from Bogdan 2025; chapter 2)
 - see the book for example sources

- Verb-medial languages:

(I) : SVO

English

(II) : OVS

Hixkaryana

- Verb-peripheral languages:

- Verb-final languages:

(III) : SOV

(IV) : OSV

Turkish (cased), Lachota (case-less)
Dyirbal

- Verb-initial languages

(V) : VSO

Welsh

(VI) : VOS

Malagasy

Clause typology by verbal category:

(5) Let's start with SVO basic order

hits :: (s \ np) / np : $\lambda x. \lambda y. \text{'hit' } x y$ English

More precisely

hits :: (s \ np_{3s}) / np : $\lambda x. \lambda y. \text{'hit' } x y$

— short for NP_{agr=3s}

There are SVO languages without agreement.

Lǐsì chén-guò yì.tiáo chuán.

Lisi

sink-ASP

one

ship

'One of Lisi's ships sank.'

Xu (2004) Mandarin Chinese

chén :: (s\NP) / NP : $\lambda x. \lambda y. \text{cause}'(\text{sink}'x \text{one}') y$
 $\wedge \text{link}'xy$
↳ no agreement

what about other options for
verb-medial languages?

hits :: (S \ NP_{3s}) / NP : $\lambda x. \lambda y. \text{hit}' \ x y$

why not :

hits :: (S / NP) \ NP_{3s} : $\lambda x. \lambda y. \text{hit}' \ y x$

BOTH are SVO !

(S/NP₃₅)/NP seems best for accusative
English because its intransitives are SV,
that is, S/NP₃₅.

Estragon hits Vladimir.

Estragon laughs.

Estragon hits Vladimir and laughs.

- A similar argument can be made for the other kind of verb-mediality: OVS

recall Hixkaryana (OVS)

yokhe ku :: $(S/NP) \backslash NP$: $\lambda x. \lambda y. \text{send}' xy$

- why not $(S \backslash NP) / NP$: $\lambda x. \lambda y. \text{verb}' yx$,
which would also be OVS?
EXERCISE!

(II) OVS languages

yokhe ku :: (S/NP) \ NP :: 7x.7y. send xy Hixkaryana
Derbyshire (1977)

Yawaka	yokhe ku	Waraka	rohya ka	oro ke
Axe	he-sent-it	W	to-me	yesterday
'Waraka sent the axe to me yesterday.'				

NB. subject and object agreement appear as affixes on the verb

Päri is ergative and verb-medial. Syntactic subject is first in the clause:

Ùbur á-pùot dháag-è.
Ubur-ABS C-beat woman-ERG
'The woman beat Ubur.'

Päri (Nilotic)

Andersen 1988:292

Ùbur á-túk'
Ubur-ABS C-play

'Ubur played.'

⇒ (S/NP)\NP seems better for Päri.

pùot :: (S/NPerg)\NPabs
: λx. λy. beat' x y

túk :: S\NPabs
: λx. play x

OVS without agreement? Probably the closest is Nadiëb (Brazil)

Salãap	ki-yúk	kalapé	hã
measles	thematic-aspect-fall-multiple	children	Native

'The children caught measles.'
Lit. 'measles fell on the children.'

Nadiëb
Weir (1986)
ex. (5a)

- Could be OSV, NB. the thematic marker in OVS. Weir says OSV.

(III): SOV languages

oku :: (S \ NP_{NOM}) \ NP_{ACC} : $\lambda x. \lambda y. \text{read}'xy$
↳ short for case = acc

Ayşe	kitabı	oku-du.	Turkish
A.NOM	book-ACC	read-PAST.3S	
'Ayşe read the book.'			

- SOV without case? Lakota (Sioux)
Dakota

Wasiču ki matho' ki φ-φ-klé.
white man the bear the 3sg-3sg-kill
_u A

the whiteman killed the bear.'

Lakhta

van Valin Jr.
(1987)

ex. (8a)

$$kte' :: (s \setminus np) \setminus np : \lambda x. \lambda y. kill' xy$$

why SOV?



(IV): OSV languages

ga bu

mother-ABS

guma-ggu

father-ERG

bura-n

see-NONFUT

Dyirbal

Dixon 1994:10

'Father saw mother.'

guma

Father-ABS

banaga-n^u

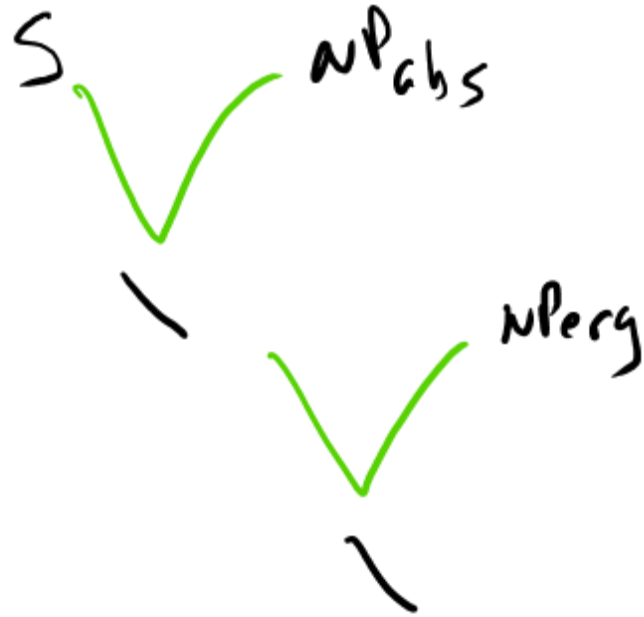
return-NONFUT

'Father returned.'

bura-n :: (s\NP_{abs})\NP_{erg}
: xy.NX.see'xy

Why is the following OSU?

bura-n :: (S \ nP_{abs}) \ nP_{erg} : $\lambda y. \lambda x. \text{see}'xy$



- Are there OSV languages without case?

- typology research suggests there is no language with

- OSV basic order

- no agreement

- no case

- Are we so sure about that?

m³³ ka⁵⁵

Muga

m³³ ko⁴⁴

Mugo

ndv²¹

beat

Liangshan Nuosu
(Southwest China)
Gerner (2004)

(1) 'Muga beat Mugo'

(2) 'Mugo beat Muga'

.... and ran away (only (1) is available)

.... and cried (only (2) is available)

IN RESULTATIVES

$ndv :: (S_{res} \setminus \mu p) \setminus \mu p : \lambda x. \lambda y. heat' y x$

$ndv :: (S \setminus \mu p) \setminus \mu p : \lambda x. \lambda y. heat' x y$

($\bar{V}$): VSO languages

Mi	newidith	Sion	ei	feddwl.
PRT	change.FUT.3s	Sion	3MS	mind.INF

'Sion will change his mind.'

Welsh

Borsley et. al 2007: 52

newidith :: $(S/np)/np_{3s}$: $\lambda y. \lambda x. \text{change}'xy$

Are there VSO ergative languages?

Wada	bube-un	na	pihu	ka	dangi=na
PERF	hit-TR	ERG	pihu	ABS	friend=3s. GEN

Seediq (Austronesian,
from Taiwan)

Aldridge 2008:986

'pihu hit his friend.'

bube-un :: (S/MP_{abs}) / ₁ERG : ₂y. ₃x. hit' xy

(VI): VOS languages

manenjika

chase

ny fiolahy

the thief

Rabe

Rabe

'Rabe is chasing the thief.'

Malagasy

(Madagascar)

Keenan 2008: 469

man-enjika :: (s/np)/np

: $\lambda x. \lambda y. \text{'chase' } xy$

we have seen that:

- all six orders are realized as basic in some language.
- Categorically, there are EIGHT possibilities, not 6!
- Syntactic command relations are free to vary (subject combining later than the object, for example.)
- Semantic Command (I-command) relations vary depending on S-command.
- All kinds of correspondences have been attested.